Iris Clustering Analysis

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Introduction

This report performs unsupervised K-means clustering on the Iris dataset to identify natural groupings of plants based on their sepal and petal measurements, as outlined in the PRD.

Data Loading and Validation

First, we load the data and validate the features.

```
file_path <- "data/iris.data"
column_names <- c("sepal_length", "sepal_width", "petal_length", "petal_width", "class")

if (file.exists(file_path)) {
  df <- read.csv(file_path, header = FALSE, col.names = column_names)
  iris_features <- df[, 1:4]
  print("Data loaded successfully.")
} else {
  stop("Data file not found.")
}
```

```
## [1] "Data loaded successfully."
```

Determining Optimal Clusters: Elbow Method

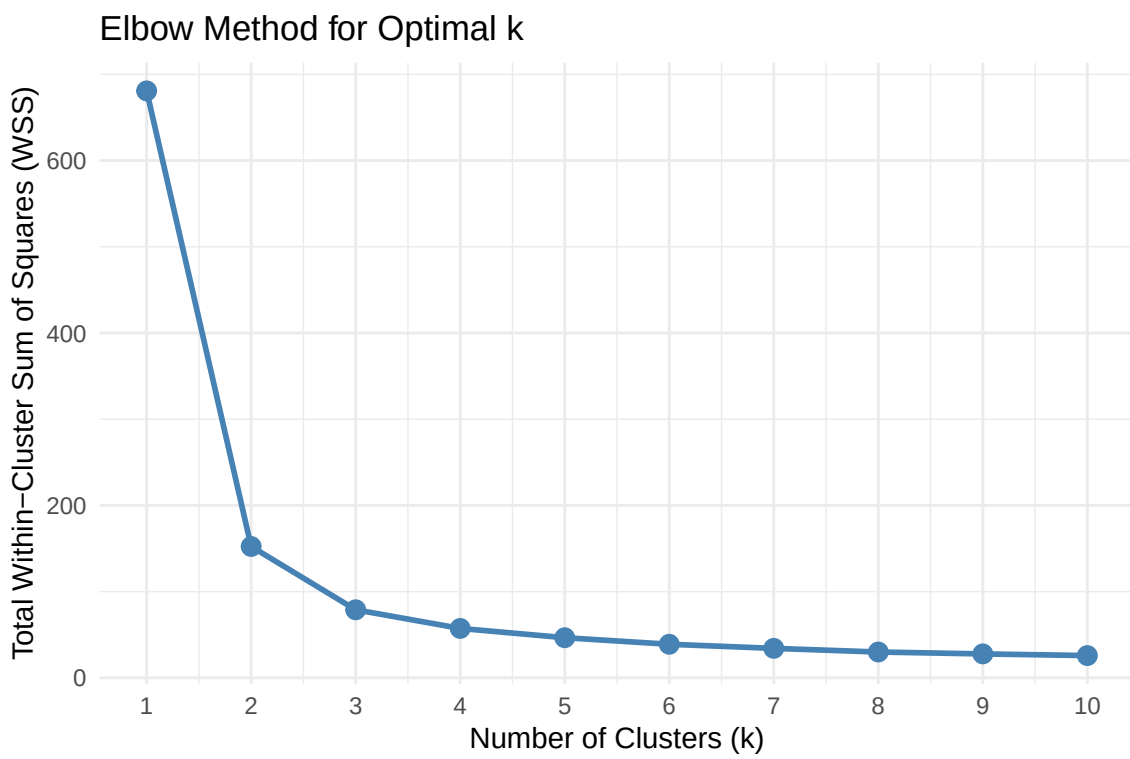
To find the optimal number of clusters (k), we evaluate the Total Within-Cluster Sum of Squares (WSS) across different cluster counts.

```
max_k <- 10
wss <- numeric(max_k)

for (k in 1:max_k) {
  set.seed(42)
  km <- kmeans(iris_features, centers = k, nstart = 20)
  wss[k] <- km$tot.withinss
}

elbow_df <- data.frame(k = 1:max_k, wss = wss)
ggplot(elbow_df, aes(x = k, y = wss)) +
  geom_line(color = "steelblue", linewidth = 1) +
```

```r
  geom_point(color = "steelblue", size = 3) +
  labs(
    title = "Elbow Method for Optimal k",
    x = "Number of Clusters (k)",
    y = "Total Within-Cluster Sum of Squares (WSS)"
  ) +
  theme_minimal() +
  scale_x_continuous(breaks = 1:max_k)
```

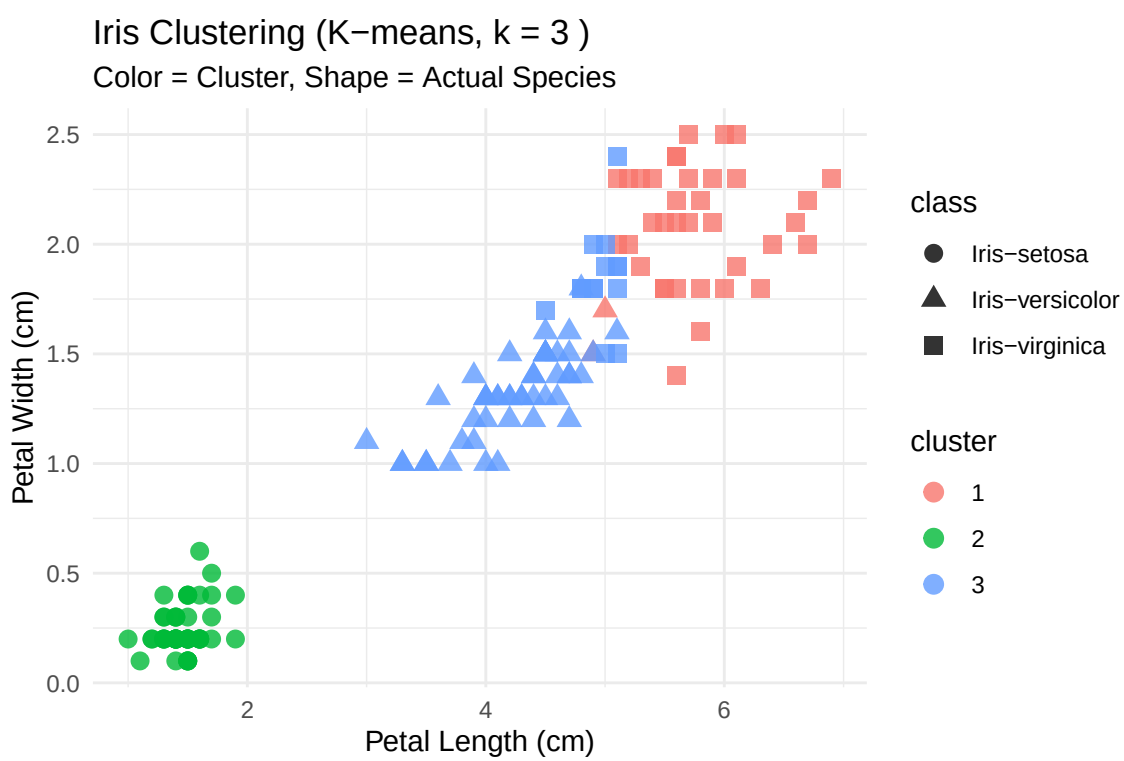


K-means Clustering Results

Based on the elbow plot, the reduction in WSS slows down significantly after $\mathbf{k = 3}$, indicating it's an appropriate choice for our number of clusters.

```r
k <- 3
set.seed(42)
iris_clusters <- kmeans(iris_features, centers = k, nstart = 20)
df$cluster <- as.factor(iris_clusters$cluster)

# Evaluate against actual labels using a Confusion Matrix
table(Cluster = df$cluster, Actual = df$class)
```

```
##         Actual
## Cluster Iris-setosa Iris-versicolor Iris-virginica
##       1           0               2            36
##       2          50               0             0
##       3           0              48            14
```

```r
# Visualization
ggplot(df, aes(x = petal_length, y = petal_width,
               color = cluster, shape = class)) +
  geom_point(size = 3, alpha = 0.8) +
  labs(
    title = paste("Iris Clustering (K-means, k =", k, ")"),
    subtitle = "Color = Cluster, Shape = Actual Species",
    x = "Petal Length (cm)",
    y = "Petal Width (cm)"
  ) +
  theme_minimal()
```